Exploratory Data Analysis

Social Media Engagement Dataset

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1. Introduction

The purpose of this exploratory data analysis (EDA) is to assess data quality, examine engagement metrics, and identify differences across social media platforms.
The analysis is descriptive in nature and aims to highlight key patterns, distributions, and limitations of the dataset.

2. Data Overview

The dataset contains ~ **10,000 social media posts**, each representing a single piece of content published on a specific platform.
For each post, engagement-related metrics are provided, including likes, comments, shares, and views, along with platform information and publication time.

The dataset covers multiple platforms and spans several months, which enables both cross-platform and basic temporal analysis.

	likes_count	shares_count	comments_count	views_count
count	9951.0	5000.0	9908.0	5621.0
mean	1414.38	79.23	299.86	17973.53
std	5989.21	979.56	1259.44	101249.23
min	0.0	0.0	0.0	3.0
25%	116.0	3.0	28.0	1539.0
50%	354.0	9.0	93.0	3704.0
75%	895.0	43.0	261.0	8535.0
max	188611.0	47500.0	80415.0	3500000.0

3. Data Quality & Missing Values

An initial inspection of the dataset revealed missing values in several engagement-related columns, most notably in `views_count` and `shares_count`.

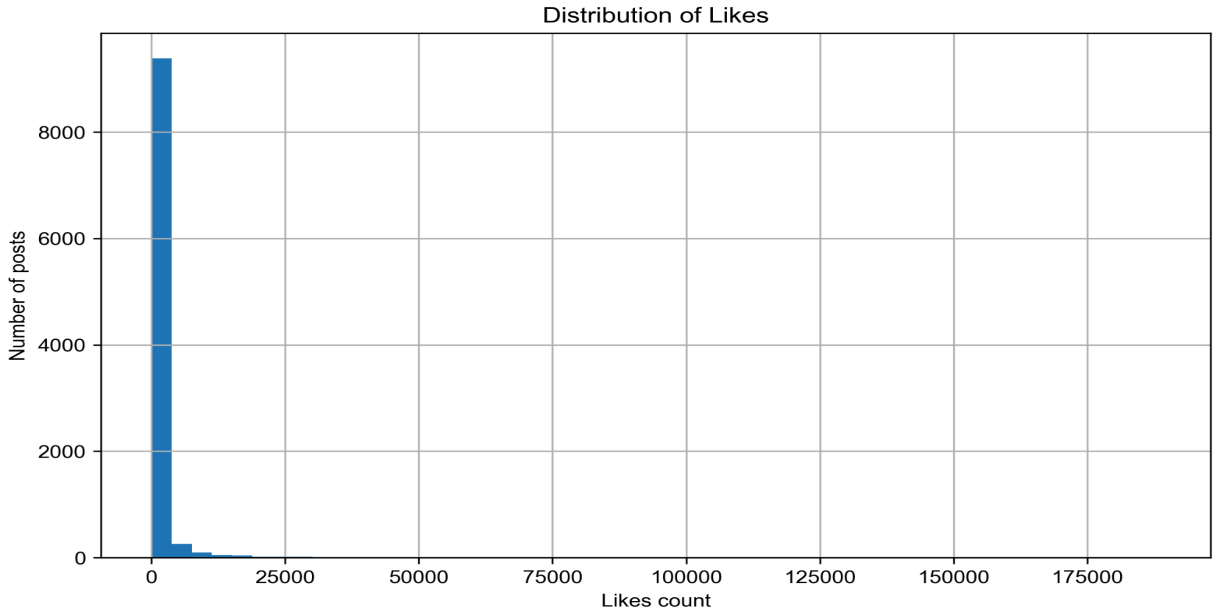
Key data cleaning steps included:

- Removing columns with no analytical value
- Converting `created_time` to a datetime format
- Calculating engagement rate only for posts with available view counts
- Preserving missing values instead of aggressive imputation to avoid distortion

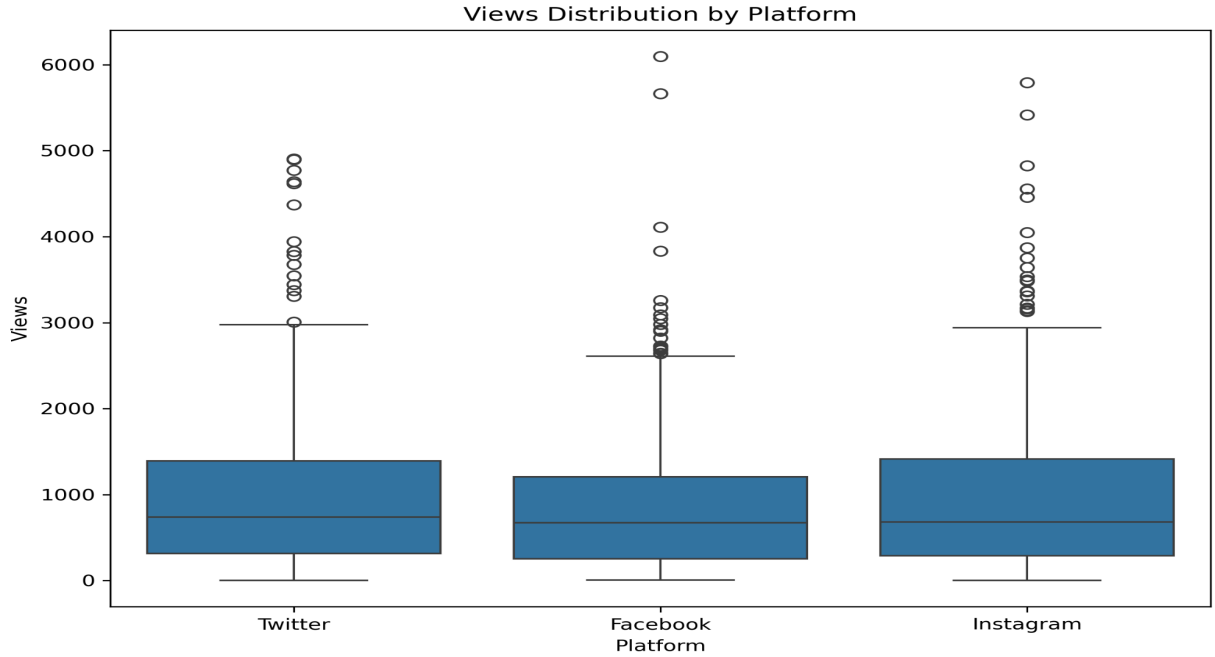
Due to missing view counts, some posts were excluded from engagement rate analysis.

4. Exploratory Visual Analysis

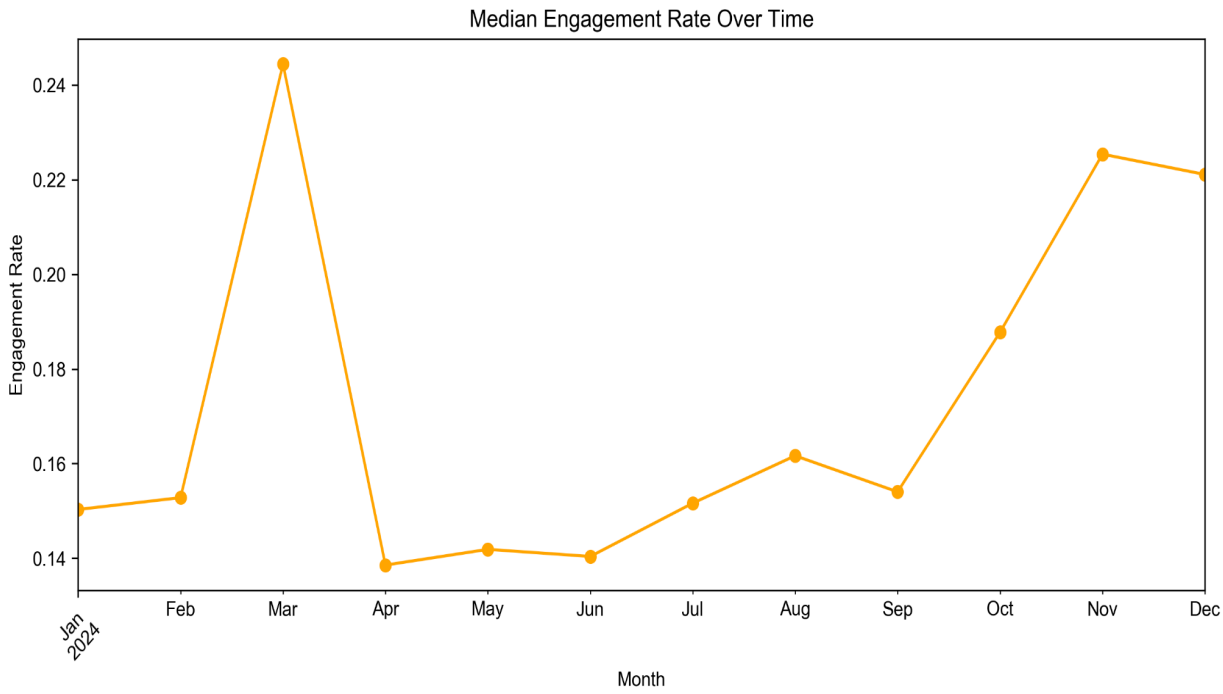
To summarize engagement patterns, several complementary visualizations were used and are presented together as a single analytical block.



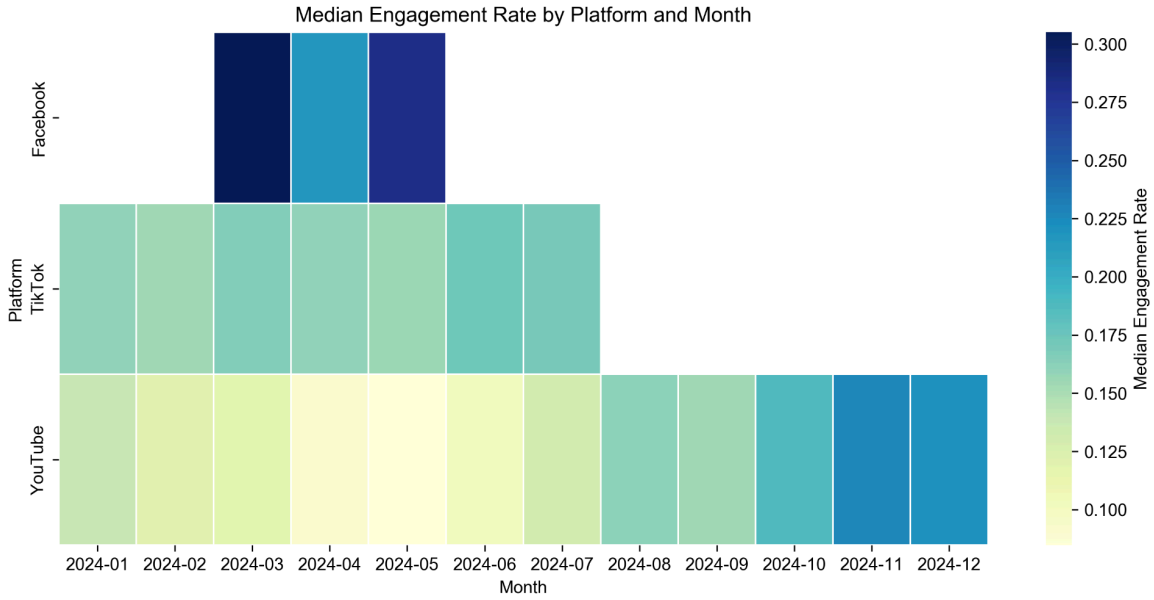
The distribution of likes is strongly right-skewed, indicating that the majority of posts receive relatively low interaction, while a small number of posts generate disproportionately high engagement. This long-tail pattern is characteristic of social media data.



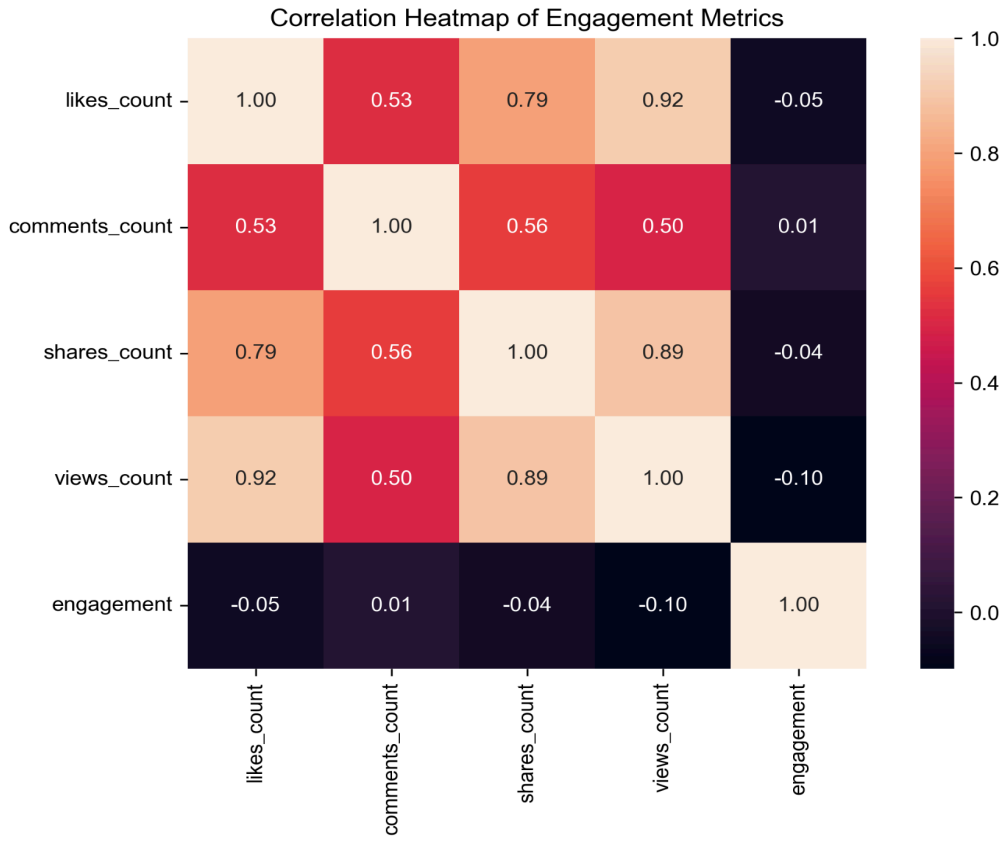
The comparison of engagement rates across platforms reveals notable differences in median engagement as well as substantial within-platform variability. The presence of extreme outliers suggests that viral content plays a significant role in overall engagement levels.



Temporal analysis of median engagement rates shows fluctuations over time rather than a stable upward or downward trend.
This suggests that engagement is driven more by content characteristics and platform dynamics than by time alone.



The platform-by-month heatmap highlights platform-specific engagement patterns and varying levels of stability over time.
Some platforms exhibit relatively consistent engagement, while others show pronounced month-to-month variation.



A correlation heatmap was used to examine relationships between engagement metrics.
Likes, comments, and shares are positively correlated.
Views show weaker correlation with other engagement metrics.
Engagement rate is moderately correlated with likes and comments, confirming its usefulness as a composite metric.

5. Key Insights

1. Engagement metrics are strongly right-skewed, with a small number of posts driving most interactions.
2. View counts contain a substantial amount of missing values, limiting engagement rate analysis.
3. Normalized engagement rate enables fairer comparison across posts with different audience sizes.
4. Engagement patterns vary significantly across platforms.
5. Temporal analysis suggests fluctuating engagement rather than a stable trend.
6. Correlation analysis confirms that likes, comments, and shares capture related but not identical user behavior.
7. Missing data may bias platform-level and time-based comparisons.

6. Limitations

- A significant portion of posts lacks view counts, restricting engagement rate calculation
- Missing shares and comments may bias engagement comparisons
- Results should be interpreted as indicative rather than definitive

Improved data completeness would significantly enhance analytical reliability.

7. Conclusion

Overall, the dataset is suitable for exploratory analysis and comparative engagement studies. However, improving data completeness—particularly for view counts—would greatly increase the robustness of platform and temporal comparisons.